

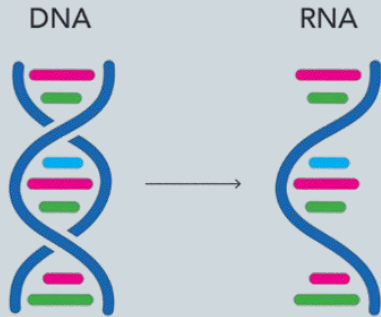
Gene Expression Based Classification of COVID-19 Using RNAseq Data

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Why this Project?

What is RNA-seq?

RNA-seq measures
how active genes are
in a cell.



Why COVID-19?

Diseases like
COVID-19 change
how genes behave.

Why my project
matters?

If fewer genes can
predict diseases, it
could simplify
diagnostics.

Introduction

- **Goal:** to determine whether gene expression data can be used to classify COVID-19 vs healthy samples, and whether a small subset of genes can accurately predict disease status.
- **Tools:** R, DESeq2, Random Forest
- **Data Cleaning:**
 - Cleaned the GEO metadata
 - Converted counts to numeric
 - Aligned samples and genes
 - Normalized with DESeq2

Background

- **Dataset**

- Main: Upper airway gene expression differentiates COVID-19 from other acute respiratory illnesses and reveals suppression of innate immune responses by SARS-CoV-2 (GSE156063)

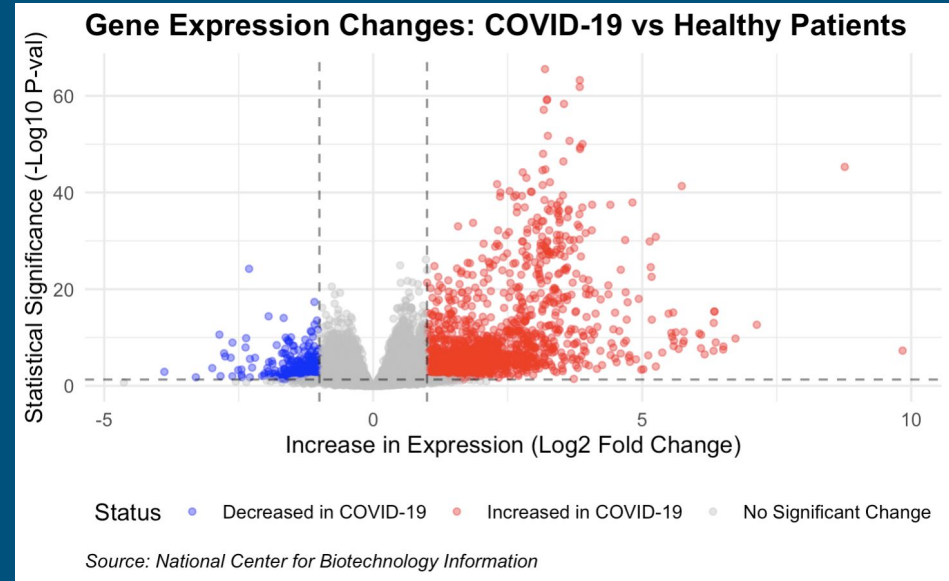
- **Key variables:**

- Innate Immune Score: measures how strongly the immune system responds to infection.
- Viral Load: measures how much virus is in the body.
- Disease Status (COVID vs. Healthy): shows whether the sample is from a COVID-19 positive or healthy person
- Gene Expression Counts: shows how active a gene is

- **Source:** public GEO dataset (NCBI)

Differential Expression

- 10,145 genes were statistically significantly ($p\text{-value} < 0.05$).
- Genes differ between COVID and Healthy.
- These genes may reflect biological differences caused by infection



Predictive Model

- Model: Random Forest
- Accuracy: 100%
- Model correctly classified all COVID-19 and Healthy samples.
 - Sensitivity = 1.0
 - Specificity = 1.0

Confusion Matrix and Statistics

Prediction	Reference	
	COVID-19	Healthy
COVID-19	16	0
Healthy	0	17

Accuracy : 1

95% CI : (0.8942, 1)

No Information Rate : 0.5152

P-Value [Acc > NIR] : 3.118e-10

Kappa : 1

McNemar's Test P-Value : NA

Sensitivity : 1.0000

Specificity : 1.0000

Pos Pred Value : 1.0000

Neg Pred Value : 1.0000

Prevalence : 0.4848

Detection Rate : 0.4848

Detection Prevalence : 0.4848

Balanced Accuracy : 1.0000

'Positive' Class : COVID-19

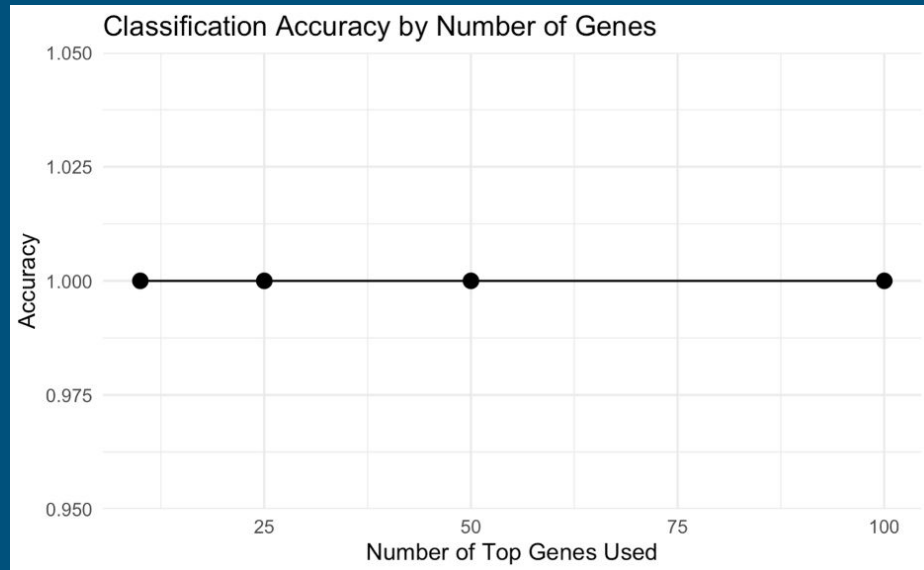
Final data Product

Gene Subset Analysis

- Tested 10, 25, 50, 100 genes
- All models achieved 100% accuracy
- Even a small subset of genes can classify disease
- More genes did not improve model performance.

X-axis = number of genes used

Y-axis = model accuracy



Key insights and Takeaways

Gene Expression Clearly separates groups

Small gene sets can classify disease

Biological + ML results align.

Findings suggest potential for simpler diagnostic approaches

Reflection

- Challenging:
 - Data Cleaning
 - Normalizing numbers
 - Combining datasets
- Rewarding
 - Building working models
 - Seeing good results
- Limitations:
 - Small sample size
 - Overfitting
 - Limited generalization

Next Steps

- Use larger datasets/subsets
- Apply cross validation
 - Would provide a more reliable estimate of model performance
 - Helps reduce overfitting
- Explore biological pathways

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- National Center for Biotechnology Information (NCBI)

References

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